

Oluwaseun Abiodun, ADEKOYA

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SUMMARY

- Mechanical Engineering PhD candidate and Robotics Engineer with 6+ years of experience in robotic systems, advanced control, manufacturing, and process optimization. Experienced in Reinforcement Learning, Model Predictive Control, CAD Design, PLC programming, DFM/DFA principles, and automation. Proven ability to develop, simulate, optimize, and evaluate PHEVs and high-DOF robotic systems for safe and efficient operation in complex environments.

EDUCATION

- Ph.D.** in Mechanical Engineering, University of Cincinnati, OH (Expected August 2027)
- MSc.** in Mechanical Engineering, University of Cincinnati, OH Dec 2024
- BEng.** in Mechanical Engineering, FUTA Akure, Nigeria **CGPA 4.73/5.00 (Summa Cum Laude)** Nov 2018
- Relevant Coursework:** Manufacturing Technologies, Machine Design, Control Systems, Industrial Engineering & Operations Research, Instrumentation & Industrial Control, Mathematical Programming

TECHNICAL SKILLS

- CAD / Design:** SolidWorks, Siemens NX, Autodesk Fusion 360, AutoCAD, Inventor | 3D Modeling, Mechanical Drawings, DFM/DFA
- Manufacturing / Process:** Lean Manufacturing, 5S, Value Stream Mapping, FMEA, OEE Analysis, SOP Development, Machining, Welding/Fabrication, 3D Printing
- Robotics / Controls:** NMPC, RL, OMPL, MoveIt, Robot Programming, PLC (Allen-Bradley), MATLAB/Simulink
- Data / Programming:** Python, PyTorch, TensorFlow, JAX, CUDA, Computer Vision, KNIME, MicroPython, Arduino, LabVIEW, FlexBatch, VirtECS
- Simulation / Robotics Software:** MuJoCo, Gazebo, Simscape, CasADi, acados, RTI Connext DDS, Sensor Integration
- Other:** Electrical schematics, Machining processes, Welding/fabrication, 3D printing, Sensor integration

PROFESSIONAL EXPERIENCE

Industry Research Engineer, Center for Intelligent Metrology and Sensing, University of Cincinnati, OH **March 2026 – Present**

- Dissertation:** Intelligent Motion Planning and Control for Safe Human-Robot Collaboration Under Motion Uncertainty in Dynamic Environments
- Developing learning-based motion planning, collision avoidance, and advanced control methods for high-DOF robotic manipulators operating in dynamic human-shared environments, using Python, MATLAB/Simulink, MuJoCo, Gazebo, and robotics simulation/control frameworks.

Intern – Production Systems and Reliability Engineering, Procter and Gamble, Mason, OH **Aug 2024 – Dec 2025**

- Built static and dynamic production system models to analyze capacity, throughput, and system reliability; identified and reduced process bottlenecks through data-driven workflow optimization.
- Collaborated with cross-functional manufacturing, reliability, and automation teams to validate system performance and implement process enhancements that improved overall equipment effectiveness.

Graduate Researcher, Integrated Vehicle Design Laboratory, UC, Cincinnati, OH **Jan 2021 – July 2024**

- Designed and implemented Nonlinear MPC algorithms in MATLAB/Simulink to optimize energy consumption and system performance for a Plug-in Hybrid Electric Vehicle powertrain using control co-design methodology.
- Led multi-year research funded by Intel, Siemens, Ethicon, John Deere, and Kinetic Vision, applying optimization and AI-driven control to Industry 4.0/5.0 robotics and manufacturing applications.

Associate Lead Engineer, Mikano International Ltd., Lagos, Nigeria **April 2019 – Dec 2020**

- Executed mechanical design, assembly, overhaul, and maintenance of generator sets (9 kVA - 2000 kVA), ensuring optimal performance, reliability, and safety compliance.
- Supervised steel fabrication, welding operations, and mechanical component integration for power generation and industrial equipment installations.
- Conducted power system audits and SOP reviews; coordinated cross-functional teams to optimize project timelines, resource allocation, and cost-effective problem resolution.

Assistant Project Engineer, British American Tobacco, Ibadan, Nigeria **Aug 2017 – Jan 2018**

- Trained in the Integrated Work System (IWS) methodology, applying lean manufacturing, sustainability, and continuous improvement principles to industrial plant operations.
- Supervised the mechanical design and installation of new piping connections for two large desiccant dryers, delivering accurate 2D CAD models to support system integration and commissioning.

PASSION PROJECTS

AI-Based Fault Detection and Anomaly Analysis for Production Line Labelers

- Pre-processed vibration sensor data using normalization and scaling techniques to ensure consistent and reliable fault analysis.
- Performed feature visualization using Matplotlib and Seaborn and Feature Selection.
- Evaluated the Data using Logistic Regression and Self-Organizing Map to distinguish faulty labelers efficiently and predict stops.

Design, Prototyping, and Optimization of a Smart Swabbing Device

- Designed mechanical components and system layouts using Fusion 360 for rapid prototyping and functional optimization.
- Programmed embedded control logic using Arduino and microprogramming techniques to enable sensor-based automation.
- Integrated multiple sensors for real-time operational feedback and device performance monitoring.
- Fabricated custom parts using 3D printing, performed soldering and electrical assembly, and assembled and tested the complete system to validate mechanical, electrical, and functional performance.

SELECTED REFEREED PUBLICATIONS

- **Adekoya, O.**, Pan, Q., Onaga, Yahia, T., K., Li, R., Dong, J., Jia, X. Critical Distance-Aware Collision Avoidance Using Probabilistic Roadmaps (PRM) and Human-Aware Motion Planners (HAMP). (Manuscript in Preparation).
- Onaga, K., Pan, Q., Yahia, T., Li, R., **Adekoya, O.**, Ma, O., Dong, J., Jia, X. Formulation and Experimental Validation of Payload-Aware Dynamic Safety Volume for Speed and Separation Monitoring. (Manuscript in Preparation).
- Yahia, T., Pan, Q., Onaga, K., Haubeil, K., **Adekoya, O.**, Ma, O., Dong, J., Jia, X. Experimental Evaluation of RGB-D Sensing for Speed and Separation Monitoring in Human-Robot Collaboration Work Cells. *Robotics and Computer-Integrated Manufacturing* (Under Review).
- Suer, A. D., Pan, Q., Yahia, T., Marasini, G., Onaga, K., **Adekoya, O.**, Haubeil, K., Dong, J., Jia, X. *Novel Methodology for 3D Vision-Guided Robotic Fastener Removal*. Measurement (Under Review).
- **Adekoya, O.**, Alexander-Ramos, M. A Comparative Study Between Dynamic Programming and Model Predictive Control for Closed-Loop Control Systems. *Journal of Mechanical Design*, 2024.
- Oluwadare, S., ElSayed, Z., **Adekoya, O.** *Unsupervised Learning Algorithms for Zero-Day Attacks in Intrusion Detection Systems*. IEEE International Conference on Computing and Machine Intelligence, 2024.
- Akinpelu, D. A., **Adekoya, O. A.**, Oladoye, P. O., Ogbaga, C. C., Okolie, J. A. *Machine Learning Applications in Biomass Pyrolysis*. Digital Chemical Engineering Journal, 2023.
- Akinboye, O. I., Zhang, Y., Krishna, V., Kondapalli, R., Olivan, L. A., Raut, N., Jackson, M., **Adekoya, O.**, White, R., Shanov, V., Bahk, J. H. *Carbon Nanotube Network-Based Composites for Enhanced Thermoelectric Performance*. arXiv Preprint.
- Alani, O. B., **Adekoya, O.**, Perrin, T., Rahman, M., Saffy, R. *Convolutional Neural Network-Based Face Mask Detection*. International Journal of Scientific and Research Publications, 2022.
- **Adekoya, O. A.**, Oluwajire, O. E., Ogunniyi, E. A. *Design and Fabrication of a Portable Solar-Powered Suction Pump*. International Journal of Scientific Research in Science, Engineering and Technology, 2019.
- Ayodeji, S. P., Gbadamosi-Adeniyi, T., Faboyin, O. T., **Adekoya, O. A.** *Design and Development of an Instant Amala Making Machine with Control System Integration*. International Journal of Scientific Research in Science, Engineering and Technology, 2019.

FUNDED RESEARCH PROJECTS

- Research on Critical distance-aware Collision Avoidance using PRMs and HAMPs, supported by Ohio Bureau of Workers Compensation (2026 - Accepted)
- Research on Comparison of Dynamic Programming and Model Predictive Control in Co-design, supported University of Cincinnati Office of Research (2024 - Accepted)
- Research on Control Co-design Methodologies for Plug-in Hybrid Electric Vehicles PowerTrain Design, supported by Industry 4.0/5.0 Grant Award with the Digital Futures (2023 - Accepted)

AWARDS AND HONOURS

- 3-Time Winner, NSBE BCA/Affiliate/Fellows Scholarship Award (2023, 2024, 2026 - Accepted)
- Paige-Lynn Roberts Memorial Scholarship (Riedman Scholarship) (2025 - Accepted)
- Graduate Student Engineer of the Month (GSEoM), CEAS, University of Cincinnati (December 2024).
- University Research Council Fellowship Award by UC Office of Research (2024 - Accepted)
- University of Cincinnati PEWS Engaging Science Fellowship Award with Gaskins Foundation (2024 - Accepted)
- NSBE R4: Tinkered Out! Scholarship Award (2024 – Accepted)
- NSF/Future of Semiconductors Fellowship with UC Center for Public Engagement with Science (2024 - Accepted)
- NSBE Academic Improvement Scholarship Award (2022 - Accepted)
- Vice chancellor award for every First-Class student in FUTA. (2018)
- Best graduating student in the Department of Mechanical Engineering, FUTA. (2018)

CERTIFICATIONS AND TRAINING

Lean Six Sigma (Yellow Belt), Analytical Hierarchy Process (AHP) Training – NSBE PI SIG, Python Essential Training, MATLAB Onramp, Simulink Onramp, Machine Learning Onramp, Introduction to Self-Driving Vehicles, Research Methods for Technical Writing